Group I, Row V: check of the derivation, UV divergences, and part 2b in dimensional regularization ($\overline{\text{MS}}$)

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1 Question and answer

Question. (1) Check the derivation of the cross section of Group I, Row V in `Group_V.tex`. (2) Does part 2a or part 2b have a UV divergence? Part 2a was treated with the plus prescription on the assumption that it is UV finite. (3) If part 2b is UV divergent, regularize it dimensionally in $\overline{\text{MS}}$; part of it should go into the running coupling, and the rest should be treated with the plus prescription as in part 2a.

Answer.

1. **The derivation is correct** (Sec. 3). We re-derived every step: the prefactor $\frac{g^4}{8\pi^4}\frac{p^+(k^+-p^+)}{k^{+3}D}$; the contraction of the $\mathbb{C}_1$ vertex with the $\mathbb{B}_2$ bracket (all nine products); the w integration; the colour algebra; the split into 2a and 2b; and the plus prescription of 2a, including its endpoint logarithms. There are only small notational points:
   - S^{ad} in the identity $f^{abc}f^{def}U^{be}U^{cf} = N_c S^{ad}$ should read U^{ad}; the identity holds for any adjoint U, symmetric or not.
   - From the w integration on, the phase $e^{-ik\cdot(z-x)p^+/k^+}$ depends on p^+ and must stand inside $\int dp^+$.

- $\int_{k^+}^0 dq^+$ with $\{[\frac{k^+}{q^+}]_+ B_2 - B_2\}$ is the same as $\int_0^{k^+} dq^+ \{-[\frac{k^+}{q^+}]_+ B_2 + B_2\}$.

2. **Part 2a has no UV divergence** (Sec. 4.2). The kernel behaves like $1/(x-z)^2$ for $x \to z$. However, the colour operator $\mathcal{O}_{2a} = [U(x') - U(w')]^{ab'}[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_c U^{af}(x)]\rho^{b'}\rho^f$ vanishes linearly at $x = z$. The same is true of the endpoint (log) terms. Your plus prescription is therefore fine.

3. **Part 2b is logarithmically UV divergent at** $x \to z$ (Sec. 4.3). Its colour operator tends to $N_c[U(x') - U(w')][U(y) - U(z)]\rho\rho$, which is the LO colour structure. The coefficient of the divergence is the gluon splitting function times LO,

$$\mathcal{V}^i \to \hat{\Phi}(\xi)\,\frac{\mathcal{K}^i(y-z)}{(x-z)^2}, \qquad \hat{\Phi}(\xi) = \xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}, \qquad \xi = \frac{p^+}{k^+}.$$

   $\hat{\Phi}$ does not depend on $d_\perp$ when the angular average is done in $d_\perp$ dimensions: the $d_\perp$ of the polarization trace cancels.

   - After your plus prescription, the endpoint terms ($\xi \to 0$ and $\xi \to 1$) carry $2\log\frac{k^+-\Lambda}{\Lambda}$ times this divergence. They are part of the JIMWLK terms, where the $1/(x-z)^2$ singularity cancels between rows.
   - The plus-prescribed remainder carries $\int_0^1 d\xi\,[\xi(1-\xi) - 2] = -\frac{11}{6} = -\frac{\beta_0}{2N_c}$. This is the running-coupling part.

4. **The $\overline{\text{MS}}$ result** (Sec. 5), with $d_\perp = 2 - \epsilon$ and $\frac{2}{\epsilon} = \frac{1}{\epsilon_{\text{std}}}$. Subtract from the plus-prescribed remainder the counterterm $[\xi(1-\xi) - 2]\,e^{i\xi k\cdot(x-z)}\mathcal{K}(y-z)/(x-z)^2$ and do its x integral in $d_\perp$ dimensions. Per ordering, with the factor 2 of the row included:

$$\boxed{\left.\frac{d^3N_{2b}}{d^3k}\right|_{\text{UV}} = \Big\{ -\frac{\alpha_s\beta_0}{4\pi}\Big[\frac{1}{\epsilon_{\text{std}}} + \log\frac{\bar{\mu}^2}{\mathbf{k}^2}\Big] - \frac{\alpha_s N_c}{4\pi}\,\frac{67}{9}\Big\}\,\frac{d^3N_{\text{LO}}}{d^3k}}$$

   Here $\mathbf{k}$ is the measured transverse momentum. The rest of 2b is finite and has the plus-prescription form of 2a (Eq. (5.9)).

5. **Running coupling** (Sec. 6). The β_0 term of Group I, Row II (part 1, the self-energy before the target) has the opposite sign, $+\frac{\alpha_s\beta_0}{4\pi}[\frac{1}{\epsilon_{\text{std}}} + \log\frac{(x-w)^2\bar{\mu}^2 e^{2\gamma_E}}{4}]$ per ordering. Per ordering, the sum of the two is free of poles and of μ, and the $\frac{67}{9}$ cancels as well:

$$\text{G1-II(1)} + \text{G1-V(2b)}\Big|_{\beta_0} = \frac{\alpha_s\beta_0}{4\pi}\log\frac{(x-w)^2\mathbf{k}^2e^{2\gamma_E}}{4} \times \text{LO} - \frac{\alpha_s N_c}{4\pi}\frac{\pi^2}{3} \times \text{LO} \quad\Longleftrightarrow\quad \alpha_s \to \alpha_s\,\frac{\alpha_s(c_0/(x-w)^2)}{\alpha_s(\mathbf{k}^2)},$$

   with $c_0 = 4e^{-2\gamma_E}$, and likewise with $x' - w'$ in the c.c. The coupling renormalization of LO itself is then not supplied by these two rows (Sec. 6.3). This is an open point; it involves the self-energy of the measured gluon after the target.

All statements were checked numerically or symbolically (Sec. 7).

2 Conventions

$\mathcal{K}(X) \equiv X/X^2$, $\mathcal{K}^i(x'-w') = (x'-w')^i/(x'-w')^2$. $\xi \equiv p^+/k^+$, $\mathbf{r} \equiv x - z$, $Y \equiv y - z$. Transverse integrals $\int_x = \int d^2x$ (in $d_\perp$ dimensions where stated). Dimensional regularization: $d_\perp = 2 - \epsilon$, so $\frac{2}{\epsilon} = \frac{1}{\epsilon_{\text{std}}}$ in the usual $D = 4 - 2\epsilon_{\text{std}}$ notation. $\overline{\text{MS}}$: every $d_\perp$-dimensional transverse integral carries $\big(\frac{\mu^2 e^{\gamma_E}}{4\pi}\big)^{\epsilon/2}$, and we write $\bar{\mu} \equiv \mu$ for the $\overline{\text{MS}}$ scale. $\beta_0 = \frac{11}{3}N_c$ ($n_f = 0$), $\alpha_s = \frac{g^2}{4\pi}$. We use the unit

$$\mathcal{N}_0 \equiv \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}. \tag{2.1}$$

LO. From `main.tex`,

$$\frac{d^3N_{\text{LO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,w,x',w'} e^{-ik\cdot(w'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\big[U(x)-U(w)\big]^{ab}\big[U(x')-U(w')\big]^{ab'}\rho^b(x)\rho^{b'}(x'). \tag{2.2}$$

Relative to its prefactor $P_{\text{LO}} = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}$ we have $\mathcal{N}_0N_c = \frac{g^2N_c}{8\pi^3}P_{\text{LO}} = \frac{\alpha_sN_c}{2\pi^2}P_{\text{LO}}$.

Your equations. We refer to the equations of `Group_V.tex` in order:

- (V.1) the row in terms of $\mathbb{A}$, $\mathbb{C}_1$, $\mathbb{B}_2$;
- (V.2) the explicit form;
- (V.3) after the w integration;
- (V.4) with the colour factor rewritten;
- (V.5) with $\pm N_c U^{af}(x)$ added;
- (V.6) part 2a with the plus prescription;
- (V.7) part 2b.

3 Check of the derivation

3.1 The ingredients

As in `Group_V.tex`:

$$-\mathbb{A}_i^a(k^+,z) = -\frac{1}{\sqrt{2}\pi}\frac{ig}{\sqrt{k^+}}\int_x \frac{(x-z)^i}{(x-z)^2}\rho^a(x), \tag{3.1}$$

$$\mathbb{C}^{bca}_{1jki}(k^+,z;p^+,x) = -\frac{gf^{abc}}{2\pi\sqrt{2k^+}}\frac{\sqrt{p^+(k^+-p^+)}}{k^+}\,\delta^{(2)}\Big(z+\frac{p^+}{k^+}x\Big)\frac{x^m}{x^2}\Big[\delta_{im}\delta_{jk}-\frac{k^+}{p^+}\delta_{jm}\delta_{ik}-\frac{k^+}{k^+-p^+}\delta_{km}\delta_{ij}\Big], \tag{3.2}$$

$$\begin{aligned}\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) = \int_x \frac{ig^2 f^{cba}\rho^a(x)\sqrt{p^+k^+}}{4\pi^2[p^+(x-z)^2+k^+(x-w)^2](p^++k^+)}\Big[&\frac{\delta^{ij}}{2(z-w)^2}\big((x-z)^2-(x-w)^2\big)\\ &+\frac{p^++k^+}{k^+}\Big(\frac{(x-z)^j(z-w)^i}{(z-w)^2}-\frac{(x-z)^j(x-w)^i}{2(x-w)^2}\Big)\\ &+\frac{p^++k^+}{p^+}\Big(\frac{(x-w)^i(z-w)^j}{(z-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2}\Big)\Big].\end{aligned} \tag{3.3}$$

In $\mathbb{C}_1$ the index j belongs to the daughter with p^+, k to the daughter with k^+-p^+, and i to the parent. In $\mathbb{B}_2$, i belongs to the gluon (k^+,w) and j to (p^+,z).

3.2 The prefactor

In Row V, $\mathbb{B}_2(k^+-p^+,-z;p^+,-x)$ emits the gluon k^+-p^+ at z (index k) and the gluon p^+ at x (index j) from the charge at y. So $k^+\to k^+-p^+$, $w\to z$, $z\to x$, $x\to y$, and $p^++k^+\to k^+$. The magnitudes are

$$|\mathbb{A}| = \frac{g}{\sqrt{2}\pi\sqrt{k^+}},\qquad |\mathbb{C}_1| = \frac{g\sqrt{p^+(k^+-p^+)}}{2\pi\sqrt{2}\sqrt{k^+}\,k^+},\qquad |\mathbb{B}_2| = \frac{g^2\sqrt{p^+(k^+-p^+)}}{4\pi^2 D\,k^+},\qquad D\equiv p^+(y-x)^2+(k^+-p^+)(y-z)^2. \tag{3.4}$$

With the factor 2 of the row,

$$2\cdot\frac{g}{\sqrt{2}\pi\sqrt{k^+}}\cdot\frac{g\sqrt{p^+(k^+-p^+)}}{2\sqrt{2}\pi\sqrt{k^+}k^+}\cdot\frac{g^2\sqrt{p^+(k^+-p^+)}}{4\pi^2 Dk^+} = \frac{g^4}{8\pi^4}\,\frac{p^+(k^+-p^+)}{k^{+3}D}, \tag{3.5}$$

as in your (V.2) (checked with `sympy`, Table 1). The factors i of $\mathbb{B}_2$ and $\mathbb{A}^\dagger$ combine to a real number. The overall sign is confirmed indirectly: the $\log(\vee/\Lambda)$ terms of this row, which come from the same expression, agree with JIMWLK⊗LO (combined note, Part III).

3.3 The contraction of $\mathbb{C}_1$ with $\mathbb{B}_2$

With j the index of the gluon at x (p^+) and k that of the gluon at z (k^+-p^+), the bracket of $\mathbb{B}_2$ is $b_1+b_2+b_3$:

$$\begin{aligned} b_1^{jk} &= \delta^{jk}\frac{(y-x)^2-(y-z)^2}{2(x-z)^2},\qquad b_2^{jk} = \frac{k^+}{k^+-p^+}\Big[\frac{(y-x)^j(x-z)^k}{(x-z)^2}-\frac{(y-x)^j(y-z)^k}{2(y-z)^2}\Big],\\ b_3^{jk} &= \frac{k^+}{p^+}\Big[\frac{(x-z)^j(y-z)^k}{(x-z)^2}+\frac{(y-x)^j(y-z)^k}{2(y-x)^2}\Big].\end{aligned} \tag{3.6}$$

The vertex of $\mathbb{C}_1^\dagger$ is $v_1+v_2+v_3 = \delta_{im}\delta_{jk}-\frac{k^+}{p^+}\delta_{jm}\delta_{ik}-\frac{k^+}{k^+-p^+}\delta_{km}\delta_{ij}$. The nine products, with $\delta^{jk}\delta^{jk}=d_\perp$:

product	result
$v_1 b_1$	$\delta_{im}\, d_\perp \frac{(y-x)^2-(y-z)^2}{2(x-z)^2} \equiv B_1^{im}$
$v_1 b_2 + v_3 b_1$	$\frac{k^+}{k^+-p^+}\delta_{im}\Big[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2(y-z)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big] \equiv \frac{k^+}{k^+-p^+}B_5^{im}$
$v_1 b_3 + v_2 b_1$	$\frac{k^+}{p^+}\delta_{im}\Big[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2(y-x)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big] \equiv \frac{k^+}{p^+}B_3^{im}$
$v_3 b_2$	$-\frac{k^{+2}}{(k^+-p^+)^2}\Big[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\Big] \equiv -\frac{k^{+2}}{(k^+-p^+)^2}B_2^{im}$
$v_2 b_3$	$-\frac{k^{+2}}{p^{+2}}\Big[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}\Big] \equiv -\frac{k^{+2}}{p^{+2}}B_4^{im}$
$v_2 b_2 + v_3 b_3$	$-\frac{k^{+2}}{p^+(k^+-p^+)}\Big[\frac{(y-x)^m(x-z)^i}{(x-z)^2} - \frac{(y-x)^m(y-z)^i}{2(y-z)^2} + \frac{(y-z)^m(x-z)^i}{(x-z)^2} + \frac{(y-x)^i(y-z)^m}{2(y-x)^2}\Big] \equiv -\frac{k^{+2}}{p^+(k^+-p^+)}B_6^{im}$

This is exactly your bracket in (V.2), checked numerically in $d_\perp = 2$ and $d_\perp = 3$ (Table 1). Multiplying by $\frac{p^+(k^+-p^+)}{k^{+2}}$ and writing $\xi = p^+/k^+$, the bracket becomes

$$\mathfrak{B}^{im}(\xi) = \xi(1-\xi)B_1 - \frac{\xi}{1-\xi}B_2 + (1-\xi)B_3 - \frac{1-\xi}{\xi}B_4 + \xi B_5 - B_6. \tag{3.7}$$

3.4 The w integration

$\delta^{(2)}\big[w - z + \xi(z-x)\big]$ sets $w = z + \xi(x-z)$: the measured gluon is at the p^+-weighted centre of the pair. Then $e^{-ik\cdot(w'-w)} = e^{-ik\cdot(w'-z)}e^{i\xi k\cdot(x-z)}$. This phase depends on p^+, so from (V.3) onwards it belongs inside $\int dp^+$. With $\int \frac{dp^+}{2\pi}\frac{1}{D} = \frac{k^+}{2\pi}\int d\xi\, \frac{1}{k^+[\xi(y-x)^2+(1-\xi)(y-z)^2]}$, the row becomes

$$\frac{d^3N_2}{d^3k} = \mathcal{N}_0 \int_{x,y,z,x',w'} \int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi\; \Psi(\xi)\, \mathcal{K}^i(x'-w')\, \mathcal{V}^i(\xi;x,y,z)\, \big[\mathcal{O}_{2a} + \mathcal{O}_{2b}\big] + \text{c.c.}, \tag{3.8}$$

$$\Psi(\xi) \equiv e^{-ik\cdot(w'-z)}e^{i\xi k\cdot(x-z)}, \qquad \mathcal{V}^i(\xi;x,y,z) \equiv \frac{(z-x)^m}{(z-x)^2}\,\frac{\mathfrak{B}^{im}(\xi)}{\xi(y-x)^2 + (1-\xi)(y-z)^2}. \tag{3.9}$$

3.5 Colour

In the first term, $f^{ebf} = -f^{fbe}$. In the second term, $f^{abc}f^{cbd} = -f^{abc}f^{dbc} = -N_c\delta^{ad}$, so $-f^{abc}f^{cbd}U^{df}(y) = +N_cU^{af}(y)$. Hence $[f^{adc}f^{ebf}U^{db}(x)U^{ce}(z) - f^{abc}f^{cbd}U^{df}(y)] = [-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_cU^{af}(y)]$, as you have.

The identity at coincident points. $f^{a'b'c'}U^{aa'}U^{bb'}U^{cc'} = f^{abc}$ for any adjoint (orthogonal) U. Multiplying by U^{gd} and using $U^TU = 1$ gives $f^{def}U^{be}U^{cf} = U^{gd}f^{gbc}$, so

$$f^{abc}f^{def}\,U^{be}(z)\,U^{cf}(z) = f^{abc}f^{gbc}\,U^{gd}(z) = N_c\,U^{ad}(z). \tag{3.10}$$

(Your S^{ad} should be U^{ad}.) With the renaming used in the row, $f^{adc}f^{fbe}U^{db}(z)U^{ce}(z) = N_cU^{af}(z)$. This holds for a general U; the symmetry $U^{bc} = U^{cb}$ is not needed.

3.6 The split and the plus prescription of 2a

Adding and subtracting $N_cU^{af}(x)$ gives

$$\mathcal{O}_{2a} \equiv \big[U(x') - U(w')\big]^{ab'}\big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_cU^{af}(x)\big]\rho^{b'}(x')\rho^f(y), \tag{3.11}$$

$$\mathcal{O}_{2b} \equiv \big[U(x') - U(w')\big]^{ab'}N_c\big[U^{af}(y) - U^{af}(x)\big]\rho^{b'}(x')\rho^f(y). \tag{3.12}$$

The kernel has $1/\xi$ (from B_4) and $1/(1-\xi)$ (from B_2). Define the endpoint kernels

$$\mathcal{V}_0^i \equiv \lim_{\xi\to 0}\xi\mathcal{V}^i = -\frac{(z-x)^mB_4^{im}}{(z-x)^2(y-z)^2}, \qquad \mathcal{V}_1^i \equiv \lim_{\xi\to 1}(1-\xi)\mathcal{V}^i = -\frac{(z-x)^mB_2^{im}}{(z-x)^2(y-x)^2}, \tag{3.13}$$

with phases $\Psi(0) = e^{-ik\cdot(w'-z)}$ and $\Psi(1) = e^{-ik\cdot(w'-x)}$. For any smooth $\mathcal{O}$,

$$\begin{aligned}\int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi\,\Psi\mathcal{V}^i = \int_0^1 d\xi\Big[\Psi(\xi)\mathcal{V}^i(\xi) - \frac{\Psi(0)\mathcal{V}_0^i}{\xi} - \frac{\Psi(1)\mathcal{V}_1^i}{1-\xi}\Big] \\ + \Big[\log\frac{\vee}{\Lambda} + \log\frac{k^+-\Lambda}{\vee}\Big]\big[\Psi(0)\mathcal{V}_0^i + \Psi(1)\mathcal{V}_1^i\big] + O\Big(\frac{\Lambda}{k^+}\Big).\end{aligned} \tag{3.14}$$

This is exactly your (V.6). There, the first integral holds $\xi(1-\xi)B_1 + (1-\xi)B_3 - [\frac{k^+}{p^+}]_+B_4 + B_4 + \xi B_5 - B_6$, i.e. $-\frac{1-\xi}{\xi} = -\frac{1}{\xi} + 1$. The second integral, in $q^+ = k^+ - p^+$, holds $-[\frac{k^+}{q^+}]_+B_2 + B_2$, i.e. $-\frac{\xi}{1-\xi} = -\frac{1}{1-\xi} + 1$. With $\mathcal{N}_0 = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$, the second line is your orange endpoint terms. The decomposition was checked numerically at fixed transverse points; the difference falls like Λ/k^+ (Table 1).

4 UV behaviour

4.1 The kernel at short distance

Put z at the origin, $x = \mathbf{r}$, $y = Y$, and let $r \to 0$. Then $\frac{(z-x)^m}{(z-x)^2} = -\frac{r^m}{r^2}$, $(y-x)^2 - (y-z)^2 = -2Y\cdot r + r^2$, and $\xi(y-x)^2 + (1-\xi)(y-z)^2 \to Y^2$. The six brackets contracted with $-r^m/r^2$ are, to leading order:

bracket	leading term of $-\frac{r^m}{r^2}B^{im}$	after the angular average in $d_\perp = d$ dims
B_1	$d_\perp \frac{r^i(Y\cdot r)}{r^4} - \frac{d_\perp}{2}\frac{r^i}{r^2}$	$\frac{d_\perp}{d}\frac{Y^i}{r^2} = \frac{Y^i}{r^2}$
B_2	$-\frac{Y^i}{r^2} + \frac{r^i}{r^2} + \frac{(Y\cdot r)Y^i}{2Y^2r^2}$	$-\frac{Y^i}{r^2}$
B_3	$-2\frac{r^i(Y\cdot r)}{r^4}$	$-\frac{2}{d}\frac{Y^i}{r^2}$
B_4	$-\frac{Y^i}{r^2} - \frac{(Y\cdot r)Y^i}{2Y^2r^2}$	$-\frac{Y^i}{r^2}$
B_5	$-2\frac{r^i(Y\cdot r)}{r^4} + 2\frac{r^i}{r^2}$	$-\frac{2}{d}\frac{Y^i}{r^2}$
B_6	$-2\frac{r^i(Y\cdot r)}{r^4} + \frac{r^i}{r^2}$	$-\frac{2}{d}\frac{Y^i}{r^2}$

We used $\langle r^i r^j\rangle = \delta^{ij}r^2/d$; terms odd in $\mathbf{r}$ average to zero. With the coefficients of Eq. (3.7),

$$\xi(1-\xi)\cdot 1 - \frac{\xi}{1-\xi}(-1) + (1-\xi)\Big(-\frac{2}{d}\Big) - \frac{1-\xi}{\xi}(-1) + \xi\Big(-\frac{2}{d}\Big) - 1\cdot\Big(-\frac{2}{d}\Big) = \hat\Phi(\xi) - \frac{2}{d}\big[(1-\xi)+\xi-1\big] = \hat\Phi(\xi), \quad (4.1)$$

so that

$$\boxed{\mathcal{V}^i(\xi;x,y,z) \xrightarrow{x\to z} \hat\Phi(\xi)\,\frac{\mathcal{K}^i(y-z)}{(x-z)^2} + \big(\text{zero angular average}\big) + O\Big(\frac{1}{|x-z|}\Big), \qquad \hat\Phi(\xi) = \xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}.} \quad (4.2)$$

$\hat\Phi = P_{gg}^{\rm real}/(2N_c)$ is the real part of the gluon splitting function. It does not depend on d:

- the $d_\perp$ of the polarization trace in B_1 cancels against the $1/d$ of the angular average;
- the $1/d$ terms of B_3, B_5, B_6 cancel among themselves for every ξ.

This is the conventional-dimensional-regularization statement that P_{gg} has no $O(\epsilon)$ part. (The earlier draft, Sec. VIII, averaged in 2 dimensions while keeping $d_\perp$ in B_1, and so got $\frac{d_\perp}{2}\xi(1-\xi)$; that should be $\xi(1-\xi)$.) The table was checked numerically in $d = 2$ and $d = 3$, and the full kernel $\mathcal{V}$ at $r = 10^{-4}$ reproduces $\hat\Phi(\xi)$ to 10^{-9} (Table 1).

4.2 Part 2a has no UV divergence

By the identity of Sec. 3.5, $\mathcal{O}_{2a}$ vanishes at $x = z$. For smooth Wilson lines it is $O(|x-z|)$ (checked: $\|\mathcal{O}_{2a}\| \propto r$, Table 1). The integrand of 2a is therefore $O(1/r)$ at small r, which is integrable in d^2r. This holds for all three pieces of Eq. (3.14) separately:

- the plus-prescribed integral;
- the endpoint terms, whose kernels $\mathcal{V}_0$, $\mathcal{V}_1$ also behave like $1/r^2$ but multiply the same $\mathcal{O}_{2a}$.

Other short-distance regions are integrable as well: $y \to x$ and $y \to z$ give $1/|y-x|$ and $1/|y-z|$, and when all three points coincide the integrand scales as r^{-3} against $d^2x\,d^2y \sim r^3dr$. Your plus prescription of 2a is therefore complete as it stands.

4.3 Part 2b is UV divergent

$\mathcal{O}_{2b} \to N_c[U(x')-U(w')]^{ab'}[U(y)-U(z)]^{af}\rho^{b'}(x')\rho^f(y)$ at $x = z$, which does not vanish. With Eq. (4.2) the integrand of 2b behaves like $\hat\Phi(\xi)/r^2$, and $\int d^2r/r^2$ diverges logarithmically. At $x = z$ we also have $\Psi(\xi) \to e^{-ik\cdot(w'-z)}$ and $1/D \to 1/(k^+Y^2)$. The divergent part is the LO cross section (gluon at z emitted by the charge at y), Eq. (2.2) with $x \to y$, $w \to z$:

$$\frac{d^3N_{2b}}{d^3k}\bigg|_{x\to z} = \frac{\mathcal{N}_0 N_c}{P_{\rm LO}}\Big[\int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi\,\hat\Phi(\xi)\Big]\int\frac{d^2r}{r^2}\times\frac{d^3N_{\rm LO}}{d^3k} = \frac{\alpha_s N_c}{2\pi^2}\Big[\int d\xi\,\hat\Phi(\xi)\Big]\int\frac{d^2r}{r^2}\times\frac{d^3N_{\rm LO}}{d^3k}. \quad (4.3)$$

The ξ integral is

$$\int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi\,\hat\Phi(\xi) = 2\log\frac{k^+}{\Lambda} - \frac{11}{6} + O\Big(\frac{\Lambda}{k^+}\Big). \quad (4.4)$$

After the plus prescription, Eq. (3.14), the two pieces are separated:

- **Endpoint terms.** $\mathcal{V}_0$ and $\mathcal{V}_1$ each tend to $\mathcal{K}(y-z)/r^2$. They carry the divergence $2\log\frac{k^+-\Lambda}{\Lambda}\times\frac{\mathcal{K}(y-z)}{r^2}$. These are the $\log(\vee/\Lambda)$ (JIMWLK) terms of the row, together with their $\log\frac{k^+-\Lambda}{\vee}$ partners. In the sum over rows the $1/(x-z)^2$ singularity cancels pointwise: every structure of JIMWLK$\otimes$LO has a colour factor that vanishes at the UV point. They need no regularization of their own, and we leave them as they are.
- **The plus-prescribed remainder.** $[1/\xi]_+$ and $[1/(1-\xi)]_+$ acting on the (ξ-independent) leading term give zero. What remains is

$$c(\xi)\,\frac{\mathcal{K}^i(y-z)}{r^2},\qquad c(\xi)\equiv\hat\Phi(\xi)-\frac{1}{\xi}-\frac{1}{1-\xi}=\xi(1-\xi)-2,\qquad \int_0^1 d\xi\,c(\xi)=-\frac{11}{6}=-\frac{\beta_0}{2N_c}. \tag{4.5}$$

  The -2 consists of the two "$+B_4$" and "$+B_2$" terms of your plus prescription. This is the running-coupling divergence, treated in Sec. 5. (Checked: $r^2\langle\text{remainder}\rangle\to-1.8333$, Table 1.)

5 Part 2b in dimensional regularization

5.1 Scheme

We use the same scheme as for the self-energy of Group I, Row II in `main.tex` ($\mathbb{A}^{(3)}$): the loop is in $d_\perp=2-\epsilon$ dimensions, the external LO factors are in 2 dimensions, and gluon polarizations are $d_\perp$-dimensional (conventional dimensional regularization; your $d_\perp$ in B_1). In $2b$ the loop variable is the size $\mathbf{r}=x-z$ of the pair, conjugate to the relative momentum of the two gluons.

5.2 The kernel in $d_\perp$ dimensions

Each factor r^m/r^2 in the pair wave functions ($\mathbb{C}_1$ and the small-r form of $\mathbb{B}_2$) is the Fourier transform of q^m/q^2. In d dimensions

$$\int\frac{d^dq}{(2\pi)^d}\,e^{iq\cdot r}\,\frac{q^m}{q^2}=\frac{i\,\Gamma(d/2)}{2\pi^{d/2}}\,\frac{r^m}{(r^2)^{d/2}}, \tag{5.1}$$

so $\frac{r^m}{r^2}\to c_d\frac{r^m}{(r^2)^{d/2}}$ with $c_d=\Gamma(\frac{d}{2})\pi^{1-d/2}$ (and $c_2=1$). The leading term $\frac{r^ir^j}{r^4}$ becomes $c_d^2\frac{r^ir^j}{(r^2)^d}$. After the angular average, $\frac{1}{r^2}\to\sigma_d(r)\equiv c_d^2\,(r^2)^{1-d}$, and $\hat\Phi$ is unchanged (Sec. 4.1).

5.3 The counterterm and its integral

We subtract from the plus-prescribed remainder its UV limit, keeping the phase $e^{i\xi k\cdot r}$ of the row as the regulator at large r:

$$\mathrm{CT}^i(\xi,\mathbf{r})\equiv c(\xi)\,e^{i\xi k\cdot r}\,\mathcal{K}^i(y-z)\,\sigma_d(r),\qquad c(\xi)=\xi(1-\xi)-2. \tag{5.2}$$

Step 1: the r integral. With $\int d^dr\,e^{iq\cdot r}(r^2)^{-a}=\pi^{d/2}2^{d-2a}\frac{\Gamma(d/2-a)}{\Gamma(a)}(q^2)^{a-d/2}$ and $a=d-1=1-\epsilon$,

$$\Big(\frac{\mu^2e^{\gamma_E}}{4\pi}\Big)^{\epsilon/2}\int d^dr\,e^{iq\cdot r}\sigma_d(r)=\pi\,\Gamma\Big(\frac{\epsilon}{2}\Big)\frac{\Gamma(1-\frac{\epsilon}{2})^2}{\Gamma(1-\epsilon)}\Big(\frac{\mu^2e^{\gamma_E}}{q^2}\Big)^{\epsilon/2}=\pi\Big[\frac{2}{\epsilon}+\log\frac{\bar\mu^2}{q^2}\Big]+O(\epsilon). \tag{5.3}$$

(Here $c_d^2=\Gamma(1-\frac{\epsilon}{2})^2\pi^\epsilon$, and the powers of π and 2 combine with the $\overline{\text{MS}}$ factor to 1. Expansion: $\Gamma(\frac{\epsilon}{2})=\frac{2}{\epsilon}-\gamma_E+O(\epsilon)$ and $\frac{\Gamma(1-\epsilon/2)^2}{\Gamma(1-\epsilon)}=1+O(\epsilon^2)$. Verified with `sympy`.) In 2 dimensions with a cutoff r_0 the same integral is $-\pi\big[\log\frac{q^2r_0^2}{4}+2\gamma_E\big]$, so $\frac{2}{\epsilon}\leftrightarrow\log\frac{4e^{-2\gamma_E}}{r_0^2\bar\mu^2}$.

Step 2: the ξ integral. With $q=\xi\mathbf{k}$,

$$\begin{aligned}\int_0^1 d\xi\,c(\xi)\,\pi\Big[\frac{2}{\epsilon}+\log\frac{\bar\mu^2}{\xi^2\mathbf{k}^2}\Big]&=\pi\Big[\Big(\frac{2}{\epsilon}+\log\frac{\bar\mu^2}{\mathbf{k}^2}\Big)\int_0^1 c(\xi)d\xi-2\int_0^1 c(\xi)\log\xi\,d\xi\Big]\\&=\pi\Big[-\frac{11}{6}\Big(\frac{2}{\epsilon}+\log\frac{\bar\mu^2}{\mathbf{k}^2}\Big)-\frac{67}{18}\Big],\end{aligned} \tag{5.4}$$

using $\int_0^1\xi\log\xi=-\frac14$, $\int_0^1\xi^2\log\xi=-\frac19$, $\int_0^1\log\xi=-1$, so $\int_0^1 c(\xi)\log\xi\,d\xi=-\frac14+\frac19+2=\frac{67}{36}$. Note that $c(0)=-2$ is finite, so $c(\xi)\log\xi$ is integrable: the $1/\xi$ parts were removed by the plus prescription before this step.

5.4 The UV part

Multiplying by $\mathcal{N}_0 N_c$ and the LO transverse structure at $x = z$, and using $\mathcal{N}_0 N_c \pi = \frac{\alpha_s N_c}{2\pi} P_{\rm LO}$:

$$\boxed{\left.\frac{d^3N_{2b}}{d^3k}\right|_{\rm UV} = \frac{\alpha_s N_c}{2\pi}\Big[-\frac{11}{6}\Big(\frac{2}{\epsilon}+\log\frac{\bar\mu^2}{\mathbf{k}^2}\Big)-\frac{67}{18}\Big]\frac{d^3N_{\rm LO}}{d^3k} = \Big\{-\frac{\alpha_s\beta_0}{4\pi}\Big[\frac{1}{\epsilon_{\rm std}}+\log\frac{\bar\mu^2}{\mathbf{k}^2}\Big]-\frac{\alpha_s N_c}{4\pi}\frac{67}{9}\Big\}\frac{d^3N_{\rm LO}}{d^3k}} \tag{5.5}$$

This is per ordering, with the factor 2 of the row included; the c.c. doubles it. $\mathbf{k}$ is the transverse momentum of the measured gluon. Explicitly,

$$\begin{aligned}\left.\frac{d^3N_{2b}}{d^3k}\right|_{\rm UV} &= \mathcal{N}_0 N_c\,\pi\Big[-\frac{11}{6}\Big(\frac{2}{\epsilon}+\log\frac{\bar\mu^2}{\mathbf{k}^2}\Big)-\frac{67}{18}\Big]\\ &\times\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\big[U(x')-U(w')\big]^{ab'}\big[U(y)-U(z)\big]^{af}\rho^{b'}(x')\rho^{f}(y).\end{aligned} \tag{5.6}$$

5.5 The finite remainder (plus prescription, as in 2a)

What is left is UV finite and is evaluated in 2 dimensions:

$$\begin{aligned}\left.\frac{d^3N_{2b}}{d^3k}\right|_{\rm fin} = \mathcal{N}_0\int_{x,y,z,x',w'}\mathcal{K}^i(x'-w')\int_0^1 d\xi\,\Big\{\Big[\Psi(\xi)\mathcal{V}^i(\xi)-\frac{\Psi(0)\mathcal{V}_0^i}{\xi}-\frac{\Psi(1)\mathcal{V}_1^i}{1-\xi}\Big]\mathcal{O}_{2b}(x)\\ -\Psi(\xi)\,c(\xi)\,\frac{\mathcal{K}^i(y-z)}{(x-z)^2}\,\mathcal{O}_{2b}(x\to z)\Big\},\end{aligned} \tag{5.7}$$

where $\mathcal{O}_{2b}(x\to z) = N_c[U(x')-U(w')]^{ab'}[U(y)-U(z)]^{af}\rho^{b'}(x')\rho^f(y)$. At small r the difference behaves as $\frac{c(\xi)\mathcal{K}(y-z)}{r^2}\big[\mathcal{O}_{2b}(x)-\mathcal{O}_{2b}(z)\big] = O(1/r)$, plus terms with zero angular average and $O(1/r)$ terms. The x integral therefore converges, with the angular integration done first. Checked numerically: $r^2\langle\cdot\rangle$ goes from -1.833 to 10^{-6} once the counterterm is subtracted (Table 1).

In your notation. The first line of Eq. (5.7) is your (V.6) with $\mathcal{O}_{2a}\to\mathcal{O}_{2b}$. The counterterm line is

$$\begin{aligned}-\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{k^+}\Big[\frac{p^+(k^+-p^+)}{k^{+2}}-2\Big]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(x-z)^2}\\ \times\big[U^{ab'}(x')-U^{ab'}(w')\big]N_c\big[U^{af}(y)-U^{af}(z)\big]\rho^{b'}(x')\rho^f(y).\end{aligned} \tag{5.8}$$

5.6 Part 2b, complete

$$\begin{aligned}\frac{d^3N_{2b}}{d^3k} &= \left.\frac{d^3N_{2b}}{d^3k}\right|_{\rm UV}+\left.\frac{d^3N_{2b}}{d^3k}\right|_{\rm fin}\\ &+\mathcal{N}_0\Big[\log\frac{\vee}{\Lambda}+\log\frac{k^+-\Lambda}{\vee}\Big]\int_{x,y,z,x',w'}\mathcal{K}^i(x'-w')\big[\Psi(0)\mathcal{V}_0^i+\Psi(1)\mathcal{V}_1^i\big]\mathcal{O}_{2b}(x)\;+\;\text{c.c.},\end{aligned} \tag{5.9}$$

with Eqs. (5.5) and (5.7). The last line is the endpoint part. As your orange terms for 2a, it reads

$$\begin{aligned}-\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\Big[\log\frac{\vee}{\Lambda}+\log\frac{k^+-\Lambda}{\vee}\Big]\int_{x,y,z,x',w'}\Big\{\\ e^{-ik\cdot(w'-z)}\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2(y-z)^2}\Big[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\Big]\\ +e^{-ik\cdot(w'-x)}\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2(y-x)^2}\Big[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\Big]\Big\}\\ \times\big[U^{ab'}(x')-U^{ab'}(w')\big]N_c\big[U^{af}(y)-U^{af}(x)\big]\rho^{b'}(x')\rho^f(y).\end{aligned} \tag{5.10}$$

5.7 Another choice of counterterm

Any counterterm with the same UV limit can be used; the finite remainder changes accordingly. With a sharp cut at a distance R instead of the phase, $\mathrm{CT}_R = c(\xi)\theta(R-r)\mathcal{K}(y-z)\sigma_d(r)$,

$$\Big(\frac{\mu^2e^{\gamma_E}}{4\pi}\Big)^{\epsilon/2}\int_{r<R}d^dr\,\sigma_d(r) = \pi\Big[\frac{2}{\epsilon}+\log\frac{R^2\bar\mu^2e^{2\gamma_E}}{4}\Big]+O(\epsilon), \tag{5.11}$$

and Eq. (5.5) becomes $-\frac{\alpha_s\beta_0}{4\pi}\big[\frac{1}{\epsilon_{\rm std}}+\log\frac{R^2\bar\mu^2e^{2\gamma_E}}{4}\big]\times$LO, without the $\frac{67}{9}$. The two choices differ by the finite 2-dimensional integral

$$\int d^2r\,\frac{e^{i\xi k\cdot r}-\theta(R-r)}{r^2}=-\pi\log\frac{\xi^2\mathbf{k}^2R^2e^{2\gamma_E}}{4}, \tag{5.12}$$

which moves between the UV part and the remainder (checked numerically). The phase is the natural regulator when $|\mathbf{k}||y-z|\gg1$: the phase then cuts the r integral at $r\sim1/(\xi|\mathbf{k}|)$ before the kernel does, and the remainder has no large logarithm. When $|\mathbf{k}||y-z|\ll1$ the kernel cuts first, at $r\sim|y-z|$, and $R=|y-z|$ is the natural choice.

6 Running coupling

6.1 The β_0 term of Group I, Row II

Part 1 of Row II (two-ρ, `Group_I.tex`) is, per ordering,

$$\begin{aligned}\frac{d^3N_{\rm II,1}}{d^3k}=\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\\ \times\Big[\Big(-\frac{2}{\epsilon}-2\gamma_E-\log\frac{(x-w)^2\bar\mu^2}{4}\Big)\Big(\frac{11}{3}+\dots\Big)-\frac{67}{9}+\frac{\pi^2}{3}+\dots\Big]\\ \times\big[U(x')-U(w')\big]^{ab'}\big[U(w)-U(x)\big]^{ab}\rho^{b'}(x')\rho^b(x),\end{aligned} \tag{6.1}$$

where the dots are rapidity logarithms. Relative to LO the prefactor is $\frac{g^2N_c}{16\pi^2}=\frac{\alpha_sN_c}{4\pi}$ and the colour factor has the opposite sign. So

$$\frac{d^3N_{\rm II,1}}{d^3k}\bigg|_{\beta_0}=\Big\{+\frac{\alpha_s\beta_0}{4\pi}\Big[\frac{1}{\epsilon_{\rm std}}+\log\frac{(x-w)^2\bar\mu^2e^{2\gamma_E}}{4}\Big]+\frac{\alpha_sN_c}{4\pi}\Big(\frac{67}{9}-\frac{\pi^2}{3}\Big)\Big\}\times\text{LO}, \tag{6.2}$$

with the logarithm inside the integral of Eq. (2.2). The logarithm in coordinate space is the Fourier transform of the momentum-space $\log(\mathbf{k}_0^2/\bar\mu^2)$ of $\mathbb{A}^{(3)}$, where $\mathbf{k}_0$ is the momentum of the gluon *before* the target.

6.2 The sum

Per ordering,

$$\boxed{\text{G1-II(1)}+\text{G1-V(2b)}\Big|_{\beta_0}=\frac{\alpha_s\beta_0}{4\pi}\log\frac{(x-w)^2\mathbf{k}^2e^{2\gamma_E}}{4}\times\text{LO}-\frac{\alpha_sN_c}{4\pi}\frac{\pi^2}{3}\times\text{LO}.} \tag{6.3}$$

- The $\frac{1}{\epsilon}$ poles cancel.
- The $\bar\mu$ dependence cancels.
- The constant $\frac{67}{9}$ cancels. It comes in both rows from $\int_0^1c(\xi)\log[\xi(1-\xi)]=2\int_0^1c(\xi)\log\xi=\frac{67}{18}$.

In terms of running couplings, $\alpha_s(\mu)\big[1+\frac{\alpha_s\beta_0}{4\pi}\log\frac{(x-w)^2\mathbf{k}^2}{c_0}\big]=\alpha_s(\mu)\frac{\alpha_s(c_0/(x-w)^2)}{\alpha_s(\mathbf{k}^2)}$ with $c_0=4e^{-2\gamma_E}$. With the c.c., $x'-w'$ appears in the second factor. The scales are physical:

- $1/|x-w|$ is the transverse scale of the emission from the projectile (before the target);
- $|\mathbf{k}|$ is that of the measured gluon (after the target).

With the cut counterterm at $R=|y-z|$ instead, the logarithms of the two rows cancel completely, and the dependence on $\mathbf{k}$ sits in the remainder of 2b. The total is the same.

6.3 Remark on the coupling renormalization

Written with the bare coupling, LO $\propto g_0^2=\mu^\epsilon g^2\big[1-\frac{\alpha_s\beta_0}{4\pi}\frac{1}{\epsilon_{\rm std}}\big]$. The NLO terms must therefore supply $+\frac{\alpha_s\beta_0}{4\pi}\frac{1}{\epsilon_{\rm std}}\times$LO in total. With the normalization of your notes, G1-II(1) gives +1 unit per ordering (+2 with the c.c.). G1-V(2b) gives −1 unit per ordering (−2 with the c.c.). They cancel. The normalization of Row V, factor 2 and c.c., is fixed independently: its $\log(\vee/\Lambda)$ terms come from the same integrand and agree with JIMWLK⊗LO. This supersedes the statement in the earlier draft (Sec. VIII) that G1-V(2b) should enter with net weight one.

The required +1 must then come from a row with a β_0 pole that is not yet included. The natural candidate is the self-energy of the measured gluon after the target: the diagonal part of $\bar{\mathbb{B}}_3^\dagger$ ("G1-I′"), which so far is known only at leading log. For an on-shell gluon this self-energy is scaleless in dimensional regularization. Its UV pole then provides the $+\frac{\alpha_s\beta_0}{4\pi}\frac{1}{\epsilon_{\rm std}}$. Its equal and opposite IR (collinear) pole is the $\frac{\beta_0}{2}\delta(1-\xi)$ term of the $\overline{\rm MS}$ gluon fragmentation counterterm. Consistently, $\alpha_s(\mu)D_g(\mu)$ is μ independent in the $\delta(1-\xi)$ sector at this order.

If this holds, the coupling structure of LO becomes $\alpha_s(c_0/(x-w)^2)\,\alpha_s(c_0/(x'-w')^2)/\alpha_s(\mathbf{k}^2)$ (at $\mu_F=|\mathbf{k}|$). This is of the "triumvirate" form found for running-coupling corrections to LO gluon production. Confirming it requires G1-I′ in dimensional regularization; we mark it as open.

7 Checks

All numbers come from `rowV_checks.py` (`rowV_checks.json`).

check	result
contraction $\mathbb{C}_1$ vertex $\times$ $\mathbb{B}_2$ bracket vs. your bracket (V.2), $d_\perp = 2, 3$, random points	max relative difference 2.7×10^{-16}
prefactor $2\lvert\mathbb{A}\rvert\lvert\mathbb{C}_1\rvert\lvert\mathbb{B}_2\rvert$ / $\frac{g^4}{8\pi^4}\frac{p^+(k^+-p^+)}{k^{+3}D}$ (`sympy`)	1
$f^{adc}f^{fbe}U^{db}U^{ce} = N_c U^{af}$ (same point; general SU(3) adjoint U)	1.3×10^{-15}
$f^{abc}f^{cbd} = -N_c\delta^{ad}$; $f^{ebf} = -f^{fbe}$	1.3×10^{-15}; 0
$r^2\langle -\frac{r^m}{r^2}B_k^{im}\rangle \cdot Y/Y^2$, $k = 1..6$, $d = 2$, $r = 10^{-4}$	1.000000, -1.000000, -1.000000, -1.000000, -1.000000, -1.000000 (expected 1.0000, -1.0000, -1.0000, -1.0000, -1.0000, -1.0000)
$r^2\langle -\frac{r^m}{r^2}B_k^{im}\rangle \cdot Y/Y^2$, $k = 1..6$, $d = 3$, $r = 10^{-4}$	1.000000, -1.000000, -0.666667, -1.000000, -0.666667, -0.666667 (expected 1.0000, -1.0000, -0.6667, -1.0000, -0.6667, -0.6667)
full kernel: $r^2\langle\mathcal{V}\rangle\, Y^2/Y$ vs. $\hat{\Phi}(\xi)$, $\xi = 0.1, 0.37, 0.8$, $r = 10^{-4}$	max relative difference 1.6×10^{-9}
$\lVert\mathcal{O}_{2a}\rVert$ at $r = 10^{-1}, 10^{-2}, 10^{-3}$ (smooth random U)	0.1147, 0.0115, 0.0012 (linear in r)
$\int_\lambda^{1-\lambda}\hat{\Phi}\, d\xi$; $\int_0^1 c(\xi)d\xi$	$-2\log\lambda - 11/6$; $-11/6$
$\int_0^1 c(\xi)\log\xi\, d\xi$; $\int_0^1 c(\xi)\log[\xi(1-\xi)]d\xi$	67/36; 67/18
$d_\perp$-dim. integrals, Eq. (5.3) and the cut version, minus the quoted $\overline{\text{MS}}$ results (`sympy`)	0; 0
$\int d^2r\,[e^{iq\cdot r} - \theta(R-r)]/r^2 = -\pi\log\frac{q^2R^2e^{2\gamma_E}}{4}$ (three (q,R))	max difference 5.4×10^{-5}
2b plus-prescribed remainder: $r^2\langle\cdot\rangle\, Y^2/Y$ at $r = 10^{-1}, 10^{-2}, 10^{-3}$	-1.8196, -1.8332, -1.8333 (UV coefficient $-\frac{11}{6}$)
same, after subtracting the counterterm: $\lvert r^2\langle\cdot\rangle\rvert$	1.1×10^{-2}, 1.1×10^{-4}, 1.1×10^{-6} (UV finite)
Eq. (3.14): direct ξ integral minus (plus integral + endpoint logs), $\Lambda/k^+ = 10^{-2}, 10^{-3}, 10^{-4}$	4.9×10^{-3}, 3.0×10^{-4}, 2.2×10^{-5} (relative), $\propto \Lambda/k^+$

Table 1: Numerical and symbolic checks.